Research appendix

Worried, but it's coming anyway · Adhurim Thaqi · W44 · 29.10.2026 · The prototype itself: https://adhurimthaqi.github.io/dataset-datastorytelling/exercises/W43_prototyping/prototype/v4.html (source: `exercises/W43_prototyping/prototype/v4.html`)

Concept refinement: what the W43 test showed

Tested: prototype "Real or AI?" (`../W43_prototyping/prototype/`), v1 → v3. **Who:** the lecturer (Test 0, v1, 22.10), my own review (v2), three testers (v3, week of 22.10). **Details:** `../W43_prototyping/prototype/test.md`

1 · What we learned

Finding	Source	Means for the concept
The **time pressure** is what people like	Lecturer (v1) + all three testers (v3)	Keep it. The clock is now part of the hook, not an extra.
It feels like **fun**, and **everything was clear**	All three testers	The hook works without explanation (acceptance 1 ✅). No tutorial needed.
It gets **predictable after several plays**: always the same four pictures	All three testers	The guess only works the first time. Replaying turns it into remembering.
The end screen was too cramped	My review (v2)	Fixed in v3: result → "Show me →" → the number.
The **57 sticks**: all three remembered it after two minutes	All three testers	The guess → number order works. Keep the pause on the 57.
Not a quiz, more a fun game challenge	All three testers	The "we are wrong if" case of the prototype card didn't happen. The game feel is a strength, so keep it.

Still to check in W48 (with 5 people and the full story):

- Does the fun game feel still hold when the "you" steps (3–4) follow, or does the change of tone feel like a break?
- Do people link the 57 to *themselves*, e.g. "I also find it hard to tell"? So far we know it wasn't a quiz and the number stuck, but nobody was asked about that link directly.

2 · What changes (concept v1 → v2)

1. **A bigger picture pool, drawn at random.**
   - at least 12 pictures (6 real, 6 AI)
   - every round draws 4: always 2 real + 2 AI, in random order
   - the same visitor doesn't get a picture twice until the pool is used up
   - **Why:** fixes the predictability. A second play is a new round, not a memory test.
2. **The hook stays short: 4 pictures × 5 seconds.**
   - The bigger pool changes *which* pictures, not *how many*. The whole experience still has to fit in about 2 minutes.
3. **The hook leads into the story.**
   - v3 ends on the 57. The full version goes on to steps 3–6 (your answer → you vs. America → the turn → the close).

    - This also tests the open hypothesis: does the guess make the number feel personal?
4. **"Play again" is offered only at the very end**, after the key message, so replaying doesn't skip the data.

3 · What stays the same

- The key message: *Most Americans are worried about AI, and most think it's coming anyway. Younger adults are more likely than older adults to think it can still be shaped.*
- The 6 steps of direction A (`../../README.md`, "Direction A")
- The data rules: latest NORC wave only, pilot excluded, "not sure / no answer" visible, no gap under the margin of error
- Neutral towards AI. The hook asks you to *look*, it doesn't say "be more critical".

User test of the prototype (v1 → v4)

From the prototype card

- **Hypothesis:** a reader who first fails (or hesitates) at the guess will feel that the real-vs-fake problem is shared, not a personal weakness.
- **We are wrong if** readers treat it as a quiz score and don't connect the number to themselves.

How (about 5 minutes per person)

1. Say one sentence only: "*This is a small prototype about AI. Please go through it and think out loud.*"
2. Hand over the phone with the newest version open (`v3.html`). **Stay silent for 3 minutes.** Don't help, don't explain.
3. Then ask:
   - "What did you take away?"
   - "What was the number at the end?" (Does the 57 stick?)
   - "Did it feel like a quiz, or like it was about you?"
   - "Was anything unclear?"
4. Two minutes later, ask again: "Which number do you still remember?"

Acceptance (from the card)

1. Understood without my help
2. Every number is on the source page
3. Surprise: something none of us expected

Notes · 3 testers, week of 22.10.2026 (v3)

	Tester 1	Tester 2	Tester 3
Who (age, AI use)	not recorded	not recorded	not recorded
Score / longest pause	not recorded	not recorded	not recorded
Said out loud while using it	not recorded	not recorded	not recorded
"What did you take away?"	positive; liked the idea	positive; liked the idea	positive; liked the idea
Number remembered (right away / after 2 min)	yes, remembered the 57	yes, remembered the 57	yes, remembered the 57
Quiz or about them?	not a quiz; a fun game challenge	not a quiz; a fun game challenge	not a quiz; a fun game challenge
Unclear	nothing, everything was clear	nothing, everything was clear	nothing, everything was clear
Surprise	not recorded	not recorded	not recorded

Summary

- **Positive overall.** All three testers liked the idea.
- **The timing works.** Testers named the time pressure as what they liked, which confirms the lecturer's comment from Test 0.

- **It's fun and clear.** It felt like fun, and all three understood it without help (acceptance point 1 ✅).
- **The number sticks.** All three still remembered the 57 two minutes later.
- **Not a quiz.** It didn't feel like a quiz at all, more like a fun game challenge. So the "we are wrong if" case on the prototype card (people treat it as a quiz score) didn't happen.
- **The issue: it gets predictable.** After playing it several times, testers knew the answers, because there are only four pictures and always the same ones. Replaying the guess loses its point.

What I change in v4 because of this:

- A bigger pool of pictures, with a random selection each round, so a second play is not a repeat.
- **Done in v4 (29.10.2026):** 16 pictures (8 real, 8 AI). Every round draws 2 real + 2 AI, with no repeats until the pool is used up (see `prompts.md`, prompt 5).

Test 0 · lecturer, in class, 22.10.2026 (v1)

- **Verdict:** "It's good."
- **Liked:** the time pressure (v1 measured the seconds and showed them after each answer).
- **Change:** make the time pressure more obvious.
- **Done in v2:** a visible 5-second countdown, a shrinking bar, and "time's up = not sure" (see `prompts.md`, prompt 3).

Review 1 · me, 22.10.2026 (v2)

- **The end screen:** the picture tiles and the 100 dots are too cramped.
- **The time:** still not obvious enough; it should be its own element, and funny.
- **Done in v3:**
  - a clock bar with lines that change every second, a burning fuse and a 3-2-1 screen
  - the end is split into "Your result" → "Show me →" → "The number" (see `prompts.md`, prompt 4)

All prompts for the prototype

Course rule (W43 slides)	"The HTML file and **all prompts** go into the research documentation."
Tool	Claude (Claude Code, model Opus 5.5), 22.10.2026
Input	Hand-drawn Crazy 6's sheet (`../photos/W43_crazy6.jpg`), box 1 "Guess", and the prototype card in `../in-class.md`
Output	`v1.html` , `v2.html` , `v3.html` : one file each, no build step, opens in any browser (needs internet for the pictures)
Structure	As taught in W43: role and goal → guardrails → experience → design reference → data, word for word → output

> **How it was made:** in the chat I asked for the prototype in one short sentence. The model already had the prototype card, the sketch and the dataset in the conversation. Prompt 1 below is the complete brief that v1 was built from, written out so that anyone can paste it into an LLM and get the same prototype.

Prompt 1: build v1

```
ROLE AND GOAL
You are a front-end developer and data-visualisation designer. Build a small, clickable prototype
of ONE moment of a data story, so that I can test it with real people this week.
Course: Data Storytelling (HSLU). Target group: young adults in Switzerland (20–35) who use AI
weekly.
The moment comes from my hand sketch (Crazy 6's, box 1 "Guess"): the reader first guesses whether
a
picture is real or made by AI, and only then sees the survey number.

HYPOTHESIS TO TEST
A reader who first fails or hesitates at the guess will feel that the real-vs-fake problem is
shared,
not a personal weakness. I am wrong if testers treat it as a quiz score and do not connect the
number
to themselves.

GUARDRAILS
- Deliver exactly one self-contained HTML file (inline CSS and JavaScript, no frameworks, no
build step).
- Phone first (390 px wide), but it must also work on a laptop. Light and dark mode.
- One aspect of the data only. Do not add other questions or numbers from the survey.
- Use only the numbers given under DATA. Do not invent, average or re-calculate anything.
  Round to whole people and show every rounding in the source line.
- Keep "not sure" and "no answer" visible as their own categories.
- No red/green, no right/wrong colours that match the colour of the data. The score stays small;
  the survey number is the hero.
- Pictures: a neutral subject (no people, no politics, no brands) with a clear licence, credited
on screen.
  All pictures show the same kind of scene, so the subject gives nothing away.
- Accessible: real buttons, keyboard shortcuts (R = real, A = AI), alt texts that do not reveal
the answer,
  respect "prefers-reduced-motion".
- Nothing is stored or sent anywhere.
```

```
EXPERIENCE (in this order)
1. Intro: title "Real or AI?" and one or two short sentences: four mountain valleys, some are
photos,
   some were made by AI, trust your gut. One "Start" button.
2. Four rounds, always in the same order (2 AI, 2 real, mixed). Each round: progress bar, one
large
   picture, two large buttons REAL / AI. After the tap: what the picture really was, where it
comes from,
   how many seconds the reader took to decide, the credit, and a "Next" button.
3. Reveal, first small: "You got X of 4", the four pictures as thumbnails marked ✓/✗, and the
reader's
   longest pause ("Your longest pause: 6.1 s on picture 3").
4. Reveal, then big: "57 of 100 Americans say they already see people around them struggling to
tell
   what's real from what's fake because of AI." Below it 100 dots (one dot = one person in 100)
fill in
   one by one, stop briefly at the 57th, then the rest fill in. Then a legend with all answers.
5. One closing line as the surprise: only 2 in 100 don't expect it at all.
6. Source line: the question word for word, the survey, the dates, n, the margin of error, the
roundings,
   and the picture credits. A "Try again" button.
Add two review shortcuts: ?step=end jumps to the reveal, ?theme=dark / ?theme=light forces a
theme.

DESIGN REFERENCE
- My course website: calm off-white surface, system font, thin grey lines, rounded cards, one
strong colour.
- Unit chart / "100 people" dot grid (as in my W38 mapping exercise): one dot = one person in
100.
- Tone of the text: short, direct, friendly, no jargon. English.

DATA (word for word; use nothing else)
Question: "Thinking about your local community, how much are you seeing each of the following
changes
because of AI? People are struggling to tell what's real from what's fake"
Subgroup: total (all US adults). Wave: 2026-09-10.
have_seen 57.13 · starting 24.39 · expect_soon 8.67 · dont_expect 1.98 · not_sure 6.94 ·
no_answer 0.89
Survey: "Americans on AI", Athena Insights with NORC at the University of Chicago,
fieldwork 10–14 Sep 2026, n = 1,083, margin of error about ±4 percentage points.

PICTURES (Wikimedia Commons)
1 AI  · "Scenic Valley in the Afternoon (Ideogram 2.0).webp" · public domain
2 Real · "2021 in Val Bavona to south.jpg" · SchmalspurDVZO, CC BY-SA 4.0
3 AI  · "Late Afternoon Cloudy Valley (SD3.5).webp" · public domain
4 Real · "Bavona river near Roseto in Val Bavona.jpg" · SchmalspurDVZO, CC BY-SA 4.0
Load them through https://commons.wikimedia.org/wiki/Special:FilePath/<file name>?width=960
and show a short message if a picture cannot load.

OUTPUT
1. The complete HTML file.
2. A list of every rounding you made, with the original value.
3. A list of everything you had to guess or decide yourself.
```

Prompt 2: corrections to v1

```
Three corrections, keep everything else the same:
1. The thumbnails of wrong guesses at the end are orange, the same colour as the 57. That mixes
up
   "you were wrong" with "people who see it". Make wrong guesses black with a ✗ instead.
2. The 100-dot grid is too big on a phone. Limit it to about 260 px wide.
3. If someone gets all four right, the heading "If that was harder than you thought…" doesn't
```

```
fit.
   In that case show: "You got all four. But it was only four pictures, and you were looking
closely."
```

Prompt 3: v2, visible time pressure (after the first test)

Test feedback (lecturer, 22.10.2026): "*It's good. I like the time pressure, but it should be more obvious.*" In v1 the time was only measured and shown afterwards. v2 makes it visible.

```
Make a copy of v1.html as v2.html and change only the time pressure. Keep everything else.

GOAL
The reader should feel the time pressure while looking at the picture, not only read about it
afterwards.
Testers said the timing was the most interesting part but it was hidden.

CHANGES
1. Intro: add one bold line "⏱ You only get 5 seconds per picture." and one line that explains
why:
   "That's about how long you look at a post before you scroll on." Add to the small print:
   "No answer in time counts as 'not sure'."
2. On each picture: a round countdown in the top-right corner (big number 5 → 1 with a ring that
empties)
   and a thin bar under the picture that shrinks. In the last 2 seconds both turn orange and the
circle pulses
   (no pulsing when the user prefers reduced motion).
3. The clock only starts when the picture has loaded, so a slow connection does not eat the time.
4. When time runs out: the buttons lock, the picture turns dark grey with "Time's up", and the
answer says
   "Time's up. That counts as 'not sure'. It was made by AI / a real photo."
5. Reveal: timed-out pictures get a ⏱ label in grey (not orange: orange is the colour of the 57).
   If the reader ran out of time at least once, replace the "longest pause" line with:
   "You ran out of time on X of 4. In the survey, 7 of 100 Americans also answered 'not sure'."
   Otherwise: "Your longest pause: X of 5 s on picture N."
6. Add the shortcut ?limit=<seconds> (2–30) so I can test other time limits.

DO NOT CHANGE
The pictures, the order, the survey numbers, the 100 dots, the source line.
```

Prompt 4: v3, a separate, playful clock and a calmer end

Feedback on v2 (my own review, 22.10.2026): the end screen is too cramped, because the picture tiles and the 100 dots sit right on top of each other. The time should be even more obvious, as its own element, and a bit funny.

```
Make a copy of v2.html as v3.html. Keep the pictures, their order, the survey numbers and the
source line.

1. TIME AS ITS OWN ELEMENT (not on top of the picture)
   - Above the picture, a full-width "clock bar": a big round number (5 → 1) on the left and a
short,
     funny line on the right that changes every second. Write one set of lines per picture so
nothing repeats,
     e.g. 5 "Take a good look 👀" · 4 "Check the light…" · 3 "Hmm. Too perfect?" ·
     2 "Your thumb wants to scroll 👆" · 1 "DECIDE!! 😬".
   - Below it, a burning fuse: a bar that shrinks with a 🔥 riding on its end.
   - In the last 40 % of the time the clock bar turns orange and the number shakes (not with
reduced motion).
   - On a tap: the clock stops and says something short ("Locked in! 🔒").
   - On time-out: the picture goes dark and blurred with "😳 Too slow. The feed moved on.",
     the clock turns grey and says "Time's up! Counts as 'not sure'."
```

```
   - Before the first picture, a separate 3-2-1 screen with huge numbers and one line each
     ("Thumbs on REAL and AI…", "Eyes open…", "Here it comes…").
   - The intro shows the rule as its own block: a round "5" and "You get 5 seconds per picture."

2. SPLIT THE END INTO TWO SCREENS, WITH A CTA BETWEEN THEM
   Screen "Your result":
   - a big score "2 / 4" and a heading that reacts:
     all right → "All four! But you knew it was a test." · at least one time-out → "The clock got
you, right?" ·
     otherwise → "Not as easy as it looks, right?"
   - the four pictures as a 2×2 grid with a label (✓ REAL / ✗ AI / ⏱ not sure) and the seconds
taken
   - small chips: ⚡ fastest, 🐢 slowest, ⏱ how often too slow
   - a bridge sentence: "You're not the only one. How many Americans already see this happening
around them?"
   - CTA button at the bottom: "Show me →"
   Screen "The number":
   - "57" very big in the data colour, "of 100 Americans" under it, then the sentence.
   - The 100 dots with more space around them (max. 320 px, 8 px gaps) and "1 dot = 1 person in
100".
   - The legend in its own card, the surprise ("Only 2 in 100 don't expect it at all") as a
callout,
     and, if the reader timed out, one line linking their time-out to the 7 of 100 who said "not
sure".
   - "Play again" button, then the source line.

3. Review shortcuts: ?step=result and ?step=number.
```

Prompt 5: v4, a random picture pool (29.10.2026)

Feedback on v3 (three testers, see `test.md`): positive, the timing works, it's fun and clear. But after several plays it gets predictable, because it's always the same four pictures. In the chat I asked: "*start v4 with the random picture pool*". The model already had the test notes and `../../W44_plan/concept-refinement.md` in the conversation. Written out as a full brief:

```
Make a copy of v3.html as v4.html. Keep the clock, the 3-2-1 screen, the result screen, the
survey numbers,
the 100 dots and the source line. Change only how the pictures are chosen.

1. A POOL INSTEAD OF FOUR FIXED PICTURES
   - At least 12 pictures, half real, half AI, all mountain valleys with a river (same scene, so
the subject
     gives nothing away). Only from Wikimedia Commons, with a licence you have checked:
     AI pictures public domain / CC0, photos CC BY or CC BY-SA, credit shown after each answer.
   - Leave out any picture with a visible watermark or AI badge, any that looks like an
illustration,
     and any with an open deletion request on Commons.
2. A RANDOM DRAW EVERY ROUND
   - Each round: exactly 2 real + 2 AI, in random order. The round is still 4 pictures × 5
seconds.
   - Remember (in localStorage, per viewer) which pictures were already shown. No repeats until
the pool
     is used up; then start over, but never repeat the round just played.
   - If storage is blocked (private mode), it must still work, just random.
   - Preload only the four pictures of the current round.
3. TEXT
   - Intro: "Two are photos. Two were made by AI." and "New pictures every round (16 in the
pool)."
   - The end button: "Play again · new pictures".
   - The source line lists the picture authors and licences.
4. Review shortcut ?reset to forget the pictures already seen.
5. Test it: 2+2 in every round, no repeat from the round before, all pictures covered before any
```

```
repeat,
   and every picture URL loads.
```

What was checked (29.10.2026):

- the draw logic in Node.js: 2,000 simulated viewers × 12 rounds. Always 2 real + 2 AI, no repeat from the round before, all 16 pictures shown within the first 4 rounds, and it still works with storage blocked.
- all 16 picture links load from Wikimedia (HTTP 200)
- licences and authors read from the Commons file pages

Prompt 6: v5, the whole story in one file (sprint 1, 29.10.2026)

In the chat I asked: "*do everything necessary and required, we want to build a good product*". The model already had the sprint plan (`../../W44_plan/documentation/02_sprint-plan.md`), v4 and the W43 AI draft in the conversation. Written out as a full brief:

```
Build v5: all six steps of direction A in ONE file, following sprint 1. Put it in
exercises/W45_sprint1/.

1. STRUCTURE
   - template.html (the page, no numbers) + build.py (reads the CSV through data_cut.py, writes
v5.html
     with every number as JSON) + check.py (compares every number in v5.html with the CSV).
2. STEPS (keep the v4 design: off-white surface, system font, rounded cards, one strong colour
per screen)
   1 Hook: the v4 Real-or-AI game, unchanged (pool of 16, 2 + 2 per round, 5-second clock).
   2 Reveal: 57 of 100 as 100 dots, legend, "only 2 in 100 don't expect it". Button "Now you →".
   3 Your answer: the sentiment question and its answer labels word for word from data.json.
   4 You vs. America: 1,000 dots + the visitor's outlined dot; tabs: all US adults and the 4 age
groups.
     Say "more" or "less" only if the gap beats the combined margin of error, otherwise "about
the same".
   5 The turn: the agency question word for word; then "2 in 3 say it's coming anyway", a stacked
bar
     with the visitor marked, and "can shape" by age as bars with the gap rule (18–34 vs 65+).
   6 Close: the key message (second sentence only while the age gap is real), the visitor's path
     (eyes / feel / think), all numbers as a table, the full source, "Play again · new pictures".
   A thin 6-step progress bar on every screen.
3. GUARDRAILS
   - No hand-typed survey number anywhere in the template. "Not sure" and "no answer" stay
visible.
   - Neutral towards AI. US adults, not Swiss opinions: say so on the last screen.
   - Nothing is saved or sent, except the list of pictures already seen (localStorage, as in v4).
   - Review shortcuts ?step=1…6, &a=, &b=, &g=, ?theme=, ?limit=, ?reset.
4. TEST
   - check.py: 0 mismatches, and it must catch a planted wrong number.
   - Click through intro → close in a browser without a script error; screenshot every step.
```

What was checked (29.10.2026):

- `check.py` : 71 checks, 0 mismatches. A planted wrong number and a hand-typed number were both caught.
- a click-through in jsdom, twice (once with two clock time-outs): every step reached, no script error
- screenshots of every step in headless Edge, light and dark mode
- Not yet: a real phone and a real tester (sprint task 5).

How the pictures were chosen

- **AI pictures:** from the Wikimedia Commons category "AI-generated landscapes".

    - From v4, all 8 come from one Commons user (VulcanSphere), who generated the same valley prompt with different models (Imagen 3.0, Reve, HiDream, AuraFlow, FLUX, Stable Diffusion 3.5). All are public domain (CC0).
- **Real photos:** from a Commons search for Swiss river valleys (Val Bavona, Ticino).
    - From v4 there are 8, by SchmalspurDVZO (CC BY-SA 4.0), Alain Rouiller (CC BY-SA 2.0) and Adrian Michael (CC BY-SA 3.0).
- **Why this scene:** every picture shows the same scene (a valley with a river), so the subject gives nothing away.
- **Left out in v4:**
    - the Ideogram 2.0 valley from v1–v3, because it has a deletion request on Commons (Sept 2026)
    - the Imagen 4.0 valley, which has a visible "ai" badge in the corner
    - the SD1.5 and FLUX poster images, which look like illustrations

Checks

All numbers checked against `exercises/W39_data/survey_all_waves_all_subgroups.csv` (see `README.md`): 0 mismatches, and the dots add up to 100.

Next prompts (v6 …)

One change per prompt, based on what testers said (`test.md`). Save each version as a new file and add the prompt here.